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1. Features

- 1-1 Vertical mounting type and small, low profile design..
- 1-2 Reflow solderable.

2. Applications

From the operating various of Mobile phones, Digital camcorder, Digital camera and Audio devices for which Low profile and Light weight designs are required.

3. Setting product

CT T - 11 35
(1) (2) (3) (4)

- (1) : Creative
- (2) : Tact switch
- (3) : Pole&Throw
- (4) : Series No.

4. Net weight : 0.035g / PCS

5. Temperature range

Parameter	Value	Unit
Operating temperature range	-20 ~ 70	℃
Preservative temperature range(Taping)	-30 ~ 80	℃

6. Standard conditions

Parameter	Value	Unit
Ambient temperature	5 ~ 35	℃
Relative humidity	45 ~ 85	% Rh
Air pressure	860 ~ 1060	mBar

If doubt arises on the design based on the measured values under the above mentioned conditions, the following conditions shall be employed.

Parameter	Value	Unit
Ambient temperature	18 ~ 22	℃
Relative humidity	60 ~ 70	% Rh
Air pressure	860 ~ 1060	mBar

7. Contact arrangement

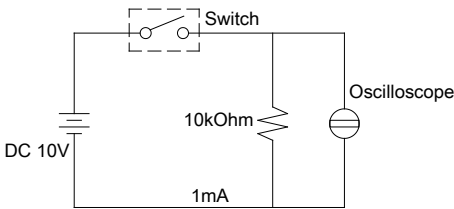
SPST

8. Type of actuating Tactile feedback

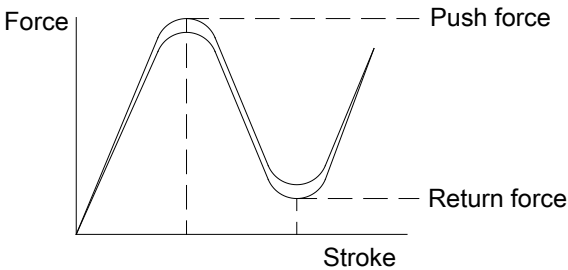
9. Rating

Maximum rating	12 V (DC) , 50 mA
Minimum rating	1 V (DC) , 10 μ A

10. Electrical characteristics

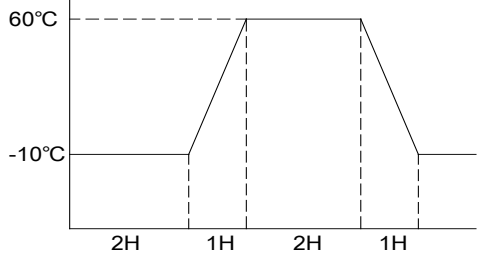
No	Item	Test conditions	Evaluation
1	Contact resistance	1-1 Push force : Operating force X 2 1-2 Small current contact resistance meter : 1Khz , 20mV , 5 ~ 50mA	100m Ω MAX.
2	Insulation resistance	100V (DC) / 1 Minute (Between terminal)	100M Ω MIN.
3	Dielectric withstand voltage	AC 250V (50Hz OR 60Hz) / 1 Minute (Between terminal)	No break down
4	Bouncing	3 to 4 Operations per second  (Switch bouncing circuit)	On / Off : 10ms MAX.

11. Mechanical characteristics

No	Item	Test conditions	Evaluation
1	Operating force	 Push by recommended operating conditions	Push force: : 160 / 220 $\pm$ 50gf Return force : 30gf MIN.
2	Travel	Operating force x 2	0.2 $\pm$ 0.1mm
3	Stem strength	1Kgf for 15 sec.	No damage (Electrical and Mechanical)

4	Soldering heat test	Mounted the switch on P.W.B by adhesive. 1) Reflow process one time. 2) Standard conditions after test : 2 Hours	No damage (Electrical and Mechanical)
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12. Environmental

No	Item	Test conditions	Evaluation
1	Storage heat	1) $+80 \pm 2^{\circ}\text{C}$, 96 Hours 2) Standard conditions after test 1 Hour.	After the test the part shall satisfy. Specifications without any degradation in appearance and performance.
2	Storage cold	1) $-30 \pm 2^{\circ}\text{C}$, 96 Hours 2) Standard conditions after test 1 Hour.	
3	Humidity	1) $40 \pm 2^{\circ}\text{C}$, 90~95% Rh , 96 Hours 2) Standard conditions after test 1 Hour.	
4	Temperature cycle	1) Test cycles : 5 Cycles 2) Standard conditions after test 1 Hour. 	

13. Endurance

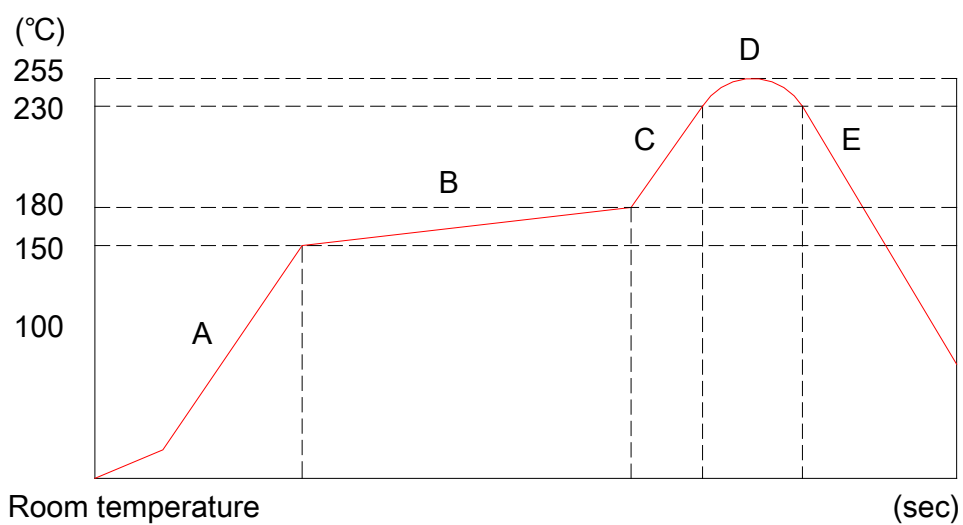
No	Item	Test conditions	Evaluation
1	Operating life	1) 5V (DC) , 50mA 2) 140 Operations/min. , Maximum value of operation force. 3) 100,000 Cycles MIN.	After the test the part shall satisfy. Specifications without any degradation in appearance and performance.

Notes

As this product is not protected from foreign material entering please make sure that any foreign material (E.G Magnetic powder, Washing solvent, Flux, Corrosive gas) Do not enter this product in your production process.

14. Solder conditions

14-1 SMD Reflow



Parts	Temperature (°C)	Time at temperature(sec)	Treatments
A	NO - 150		
B	150 - 180	90 ± 30	Pre-heating zone
C	180 - 230		
D	230 – 255 - 230	30 ± 10 (Peak : 3 MAX.)	Soldering zone
E	230 to NO		Cooling zone

** NO : Normal conditions

14-2 Solder cream thickness

0.12 ± 0.03

14-3 Manual soldering

14-3-1 Iron tip temperature : 350 ± 5°C MAX.

14-3-2 Duration : 3sec MAX.

15. Dimensions & Material list

